
UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

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ABSTRACT

Two fundamental families of explanations for machine-learning predictions have anchored the study of explanation complexity: (i) *sufficient reasons*, which are subsets of input features that, once fixed, determine a prediction, and (ii) *contrastive reasons*, which are subsets of input features that, once modified, alter a prediction. These explanations have hitherto been analyzed in isolation, dispersed across mutually disconnected formulations — non-probabilistic versus probabilistic, and local versus global — with no principled account of the structure they share. We close this gap decisively, introducing a single framework that subsumes every one of these variants by recasting each, exactly, as the minimization of one common probabilistic value function. Within this framework we establish that the complexity of every such computation is governed by precisely three structural properties of the value function: (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity* — the three cornerstones of *combinatorial optimization*. Through this lens the local and global regimes are shown to obey a sharp and counterintuitive dichotomy: the *local* value functions possess none of these properties — neither monotonicity nor submodularity/supermodularity — whereas we prove that their *global* counterparts possess all three. This dichotomy is the analytical engine of our results. Exploiting it, we derive a series of new polynomial-time algorithms that compute global explanations with provable guarantees across the full interpretability spectrum of model classes — neural networks, decision trees, and tree ensembles — in exactly the regime where the corresponding local computations are intractable: we prove that even drastically simplified instances of these same explanations are NP-hard locally, thereby delineating the precise tractability frontier that separates the two regimes.

1 INTRODUCTION

Despite substantial progress in methods for explaining the decisions of machine-learning (ML) models, the literature has converged on a robust and sobering conclusion: the most desirable explanation types — precisely those equipped with provable guarantees — are computationally hard to obtain (Barceló et al., 2020; Van den Broeck et al., 2022), and this hardness only intensifies for the complex, highly non-linear models on which reliable explanations are most urgently needed (Barceló et al., 2020; Adolphi et al., 2025; 2024). The *computational complexity* of obtaining explanations has consequently become a central theoretical concern, driving a sustained effort to chart, for each model class, exactly which explanation types are efficiently computable and which remain intractable (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022). Yet this effort has proceeded piecemeal: each explanation type has been scrutinized within its own isolated setting, so that the field still lacks a principled account of what governs tractability across the landscape as a whole. Supplying precisely such an account — and demonstrating its algorithmic payoff — is the goal of the present work.

From sufficient to contrastive reasons. Two explanation types have attracted the most sustained attention in the study of explanation complexity, and they anchor our analysis: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a subset of input features S that, once fixed to specific values,

determines the model’s prediction irrespective of every assignment to the complementary set $\bar{S}$. Its exact dual, a *contrastive reason*, is a subset of input features S whose modification alone suffices to overturn the prediction.

Additive attribution methods distribute importance scores across all features, but the resulting scores are frequently difficult for humans to interpret (Kumar et al., 2020) and afford limited actionability (Bilodeau et al., 2024). Sufficient and contrastive reasons instead furnish *discrete, condition-like explanations* that address directly the two questions practitioners actually pose: “what is enough to justify this prediction?” and “what must change to overturn it?”. This discrete, human-aligned character is precisely what underlies their central role across a broad range of classical XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), their documented superiority over additive models at improving human predictive accuracy (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and their established utility for downstream tasks such as bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). A further well-established principle holds that *smaller* sufficient and contrastive explanations are more interpretable, thereby elevating *minimality* to a central guarantee of interest in its own right (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

One of the earliest complexity-theoretic studies of sufficient and contrastive reasons, due to Barceló et al. (2020), already reveals how sharply tractability depends jointly on the model class and the explanation type. Computing a minimum-cardinality sufficient reason is NP-hard for decision trees and escalates to Σ_2^P -hardness for neural networks; for contrastive reasons the picture inverts, the minimum-size explanation being polynomial-time computable for decision trees yet again NP-hard for neural networks. These hardness results were subsequently extended to tree ensembles (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b). Collectively, they demonstrate that neither the model class nor the explanation type alone determines tractability — a pattern that our framework ultimately resolves along a single structural axis.

From non-probabilistic to probabilistic explanations. The classical definitions of sufficient and contrastive reasons are widely criticized for their rigidity: by insisting that a guarantee hold in an absolute sense across the entire domain, they routinely yield explanations that are excessively large and, in practice, uninformative (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). To overcome this rigidity, the field has converged on a strictly more general *probabilistic* formulation of sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023), which seeks subsets of input features that determine or alter a prediction with probability exceeding a prescribed threshold δ .

Wäldchen et al. (2021) initiated the complexity-theoretic study of probabilistic sufficient reasons, proving the problem NP^{PP}-hard already for CNF classifiers, with the hardness propagating to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). The pivotal theoretical insight of the probabilistic setting is the *failure of monotonicity* in the underlying probability function (Arenas et al., 2022; Izza et al., 2023) — a failure that renders even *subset-minimal* explanations computationally hard. Strikingly, (Arenas et al., 2022) establish that computing a subset-minimal probabilistic sufficient reason is NP-hard even for decision trees, in sharp contrast to the non-probabilistic case, in which the identical task admits a polynomial-time algorithm (Huang et al., 2021; Barceló et al., 2020). Non-monotonicity thus emerges as the precise structural obstruction that separates tractable from intractable explanation problems — and it is exactly this obstruction that our framework isolates and, in the global regime, turns to decisive advantage.

From local to global explanations. More recently, Bassan et al. (2024) broadened the complexity analysis from the *local* (non-probabilistic) setting — where an explanation is tied to a single prediction — to the *global* (non-probabilistic) setting, which seeks sufficient or contrastive reasons valid across the entire domain. As with other non-probabilistic formulations (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), however, the resulting criteria are exceedingly stringent — arguably more so globally than locally: any feature excluded from a global sufficient reason is deemed strictly redundant (Bassan et al., 2024), so the explanation typically absorbs nearly all input features and thereby forfeits its informativeness. This tension frames the central question of our work: does the probabilistic relaxation, which tempers the strictness of local explanations, reshape the global landscape as well — and, if so, in whose favor? We answer this question in

full, and the answer is unambiguous: the relaxation transforms the global regime into one that is not merely well-behaved but algorithmically tractable, and does so exactly where the local regime remains hard.

OUR CONTRIBUTIONS

1. **A unifying framework.** We unify the previously fragmented landscape of explanation-computation problems — sufficient and contrastive, probabilistic and non-probabilistic, local and global — within a single framework that recasts every variant as the minimization of one common value function. We then isolate exactly the three structural properties of this value function that dictate the complexity of the task: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity*. The effect is to collapse a diverse collection of complexity questions onto the analysis of a single object.
2. **A local–global structural dichotomy.** We prove that these properties behave in *diametrically opposed* ways across the local and global regimes. The *local* value functions for both sufficient and contrastive reasons are *non-monotonic*; yet we establish the striking result that their *global* counterparts are *monotonic non-decreasing*. We further uncover structure unique to the global regime: the global sufficient value function is *supermodular* and the global contrastive value function is *submodular*, whereas locally neither property holds. This dichotomy is the analytical engine that drives every one of our tractability results.
3. **Polynomial-time subset-minimal global explanations.** Exploiting this structure, we prove that global explanations with provable guarantees are efficiently computable exactly where their local counterparts are intractable, and we do so across three model classes spanning the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles. Concretely, whereas computing a subset-minimal local probabilistic sufficient/contrastive reason is NP-hard even for decision trees (Arenas et al., 2022), we prove that its global counterpart is computable in polynomial time. We then extend this positive result to *any* model admitting polynomial-time inference — including complex black-box models such as neural networks and tree ensembles — under empirical distributions, where the subset-minimal global sufficient/contrastive explanation is obtainable in polynomial time even as the local version provably remains NP-hard.
4. **Constant-factor approximations for cardinally minimal global explanations.** We establish a still stronger separation by exploiting the submodularity and supermodularity that hold globally but fail locally. For the harder task of computing *cardinally minimal* global explanations, we obtain provable *constant-factor* approximation guarantees — even for neural networks and tree ensembles — whenever the empirical distribution is fixed. In sharp contrast, we prove strong inapproximability in the local setting: no *bounded* approximation is achievable there, even under drastically simplified assumptions. The local–global divide therefore persists undiminished from exact computation into the approximation regime.

Owing to space constraints, the main text states our principal theorems and corollaries, with complete proofs deferred to the appendix.

2 PRELIMINARIES

Setting. We consider an input space of dimension $n \in \mathbb{N}$, whose input vectors $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ have each coordinate $\mathbf{x}_i$ ranging over a domain $\mathcal{X}_i$ that may be either discrete or continuous. The full input feature space is accordingly the product $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$, and we analyze classification models $f : \mathbb{F} \rightarrow [c]$ over $c \in \mathbb{N}$ classes. Explanations are evaluated with respect to a *generic* distribution $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$ over the input space. Since faithfully approximating $\mathcal{D}$ is computationally infeasible in many settings, the standard and practical alternative is to operate over a fixed empirical dataset $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$ that serves as a proxy for the “true” distribution — for instance, a sampled subset of the available training data.

The explanations we study are of two kinds, local and global. A *local* explanation targets a specific prediction $\mathbf{x} \in \mathbb{F}$, articulating why the model returned $f(\mathbf{x})$ on that particular instance. A *global* explanation, by contrast, abstracts away from any single input and characterizes the decision-making logic that governs the behavior of f across the input space in its entirety.

Models. Although many of our results are fully general — most notably the intrinsic properties of the value functions — we complement them with model-specific complexity results for widely deployed ML models. To span the interpretability spectrum in its entirety, we analyze model classes ranging from those regarded as “black-box” to those regarded as “interpretable”: (i) decision trees, (ii) neural networks (encompassing every architecture at least as expressive as a feed-forward ReLU network), and (iii) tree ensembles, covering both majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). This selection is deliberate: it ensures that our tractability guarantees reach the most expressive model classes in common use, not merely the structurally simple ones. Formal definitions are provided in the Appendix.

Distributions. In probabilistic explanation settings the complexity depends critically on the input distribution $\mathcal{D}$, and we therefore delineate three distribution classes that jointly cover the assumptions invoked throughout our analysis: (i) *general distributions*, which impose no assumption whatsoever on $\mathcal{D}$ and thus subsume every possible distribution — these underpin our universally valid structural results; (ii) *empirical distributions*, comprising all distributions induced by a finite dataset $\mathbf{D}$, the workhorse assumption of practical XAI (Lundberg & Lee, 2017; Van den Broeck et al., 2022); and (iii) *independent distributions*, which posit mutual independence among the features of $\mathcal{D}$ — a further assumption in widespread use (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). Crucially, empirical distributions do *not* presuppose feature independence; on the contrary, they encode the rich dependency structure extracted from finite datasets (Van den Broeck et al., 2022), and it is precisely this expressiveness that renders our empirical-distribution tractability results applicable to realistic modeling regimes rather than to idealized special cases. Complete formal definitions are deferred to the Appendix.

3 FORMS OF REASONS

This section formalizes the explanation types studied throughout the paper. We begin with the strict non-probabilistic definitions of (local and global) sufficient and contrastive reasons, and then generalize each to its more flexible probabilistic counterpart, from which the non-probabilistic definition is recovered exactly as the limiting case $\delta = 1$.

3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

Sufficient reasons. A central goal of *feature selection* is to identify the subset of features responsible for a model’s decision, and the *sufficiency* criterion — adopted by a range of established explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022) — renders this notion precise. Sufficiency admits both a local instantiation, anchored to a single prediction, and a global one, quantified over the entire input domain. Following standard conventions, a *local sufficient reason* is a subset of features $S \subseteq \{1, \dots, n\}$ for which fixing the features in S to their values in $\mathbf{x}$ preserves the prediction $f(\mathbf{x})$ under *every* assignment to the complementary features $\bar{S}$. Formally, S is a local sufficient reason for $\langle f, \mathbf{x} \rangle$ iff:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes the vector whose features in S are inherited from $\mathbf{x}$ and whose features in $\bar{S}$ are inherited from $\mathbf{z}$. Local sufficient reasons are intimately connected to *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which seek an alternative input $\mathbf{z}'$ that preserves the prediction (i.e., $f(\mathbf{z}') = f(\mathbf{x})$) while remaining as “close” as possible to the original point: instantiating $\mathbf{z}'$ as $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ and measuring proximity by Hamming distance renders the two notions identical, so that our results transfer to that setting without modification.

A *global sufficient reason* (Bassan et al., 2024) strengthens this criterion, demanding a subset of input features $S \subseteq \{1, \dots, n\}$ that constitutes a local sufficient reason simultaneously for *every* input $\mathbf{x} \in \mathbb{F}$:

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

Contrastive reasons. A complementary and equally prevalent paradigm explains a prediction by identifying subsets of input features whose *modification* alters it (Dhurandhar et al., 2018; Mothilal

et al., 2020; Guidotti, 2024), thereby isolating the minimal change required to overturn the decision. Formally, a subset $S \subseteq \{1, \dots, n\}$ is a *local contrastive reason* for $\langle f, \mathbf{x} \rangle$ iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Contrastive reasons are, in turn, tightly linked to *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020), which seek a nearby assignment $\mathbf{x}'$ that flips the prediction, i.e., $f(\mathbf{x}') \neq f(\mathbf{x})$; under the Hamming metric the two notions again coincide, so that our analysis extends to that setting as well.

Mirroring the global sufficiency criterion, one may instead demand a subset of input features whose alteration flips the prediction for *all* inputs in the domain of interest. This global contrastive notion connects directly to techniques for detecting bias (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020) and to *group* counterfactual methods that seek counterfactuals across many instances simultaneously (Carrizosa et al., 2024; Warren et al., 2024). Formally, S is a *global contrastive reason* for f iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As motivated in the introduction, the non-probabilistic definitions impose conditions too stringent for practical use, which is why the field has embraced strictly more general *probabilistic* definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021): rather than requiring a guarantee to hold absolutely, they require it to hold with probability at least $\delta \in [0, 1]$. Setting $\delta = 1$ recovers the original non-probabilistic definitions exactly, so the probabilistic formulation constitutes a genuine generalization rather than a departure — and every result we establish for it specializes, without loss, back to the classical case.

Probabilistic sufficient reasons. A subset $S \subseteq \{1, \dots, n\}$ is a *local δ -sufficient reason* for $\langle f, \mathbf{x} \rangle$ when fixing the features in S to their values in $\mathbf{x}$ and drawing the remaining features from $\mathcal{D}$ preserves the classification with probability at least δ ; that is:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where $\mathbf{z}_S = \mathbf{x}_S$ indicates that the features in S of $\mathbf{z}$ are fixed to their values in $\mathbf{x}$. Following the standard construction of global explanations by averaging over the input domain, we call a subset $S \subseteq \{1, \dots, n\}$ a *global δ -sufficient reason* for $\langle f \rangle$ when the expectation of the local sufficiency probability, taken over inputs $\mathbf{x}$ drawn from $\mathcal{D}$, is at least δ ; that is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

Probabilistic contrastive reasons. Dually, a *local δ -contrastive reason* is a subset of input features whose variation flips the prediction with sufficient probability. In contrast to the sufficient case, here the *complementary* features $\bar{S}$ are held fixed to their values in $\mathbf{x}$ while the features in S are permitted to vary; S qualifies precisely when the prediction differs from $f(\mathbf{x})$ with probability at least δ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

In the global setting, and in exact analogy with global sufficient reasons, a subset S is a *global δ -contrastive reason* precisely when the expectation of the local contrastive probability, taken over inputs $\mathbf{x}$ drawn from $\mathcal{D}$, is at least δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

4 FORMS OF MINIMALITY

As noted in the introduction, across every explanation form introduced so far — sufficient or contrastive, local or global — the literature converges on the principle that *smaller* explanations are

more meaningful, thereby elevating *minimality* to a provable guarantee of primary interest. We study two central and complementary notions of minimality, both of which apply uniformly to every one of our explanation types:

Definition 1. Assuming a subset $S \subseteq \{1, \dots, n\}$ is an explanation, then:

1. S is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff S has the smallest explanation cardinality $|S|$ (i.e., there is no explanation S' such that $|S'| < |S|$).
2. $S \subseteq \{1, \dots, n\}$ is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff S is an explanation, and any $S' \subsetneq S$ is not an explanation.

Cardinal minimality is strictly stronger than subset minimality: every cardinally minimal S is subset-minimal, whereas the converse fails (Appendix B.3 supplies an explicit example). We adopt the terms *subset* and *cardinally* minimal, in preference to local and global minima, precisely to forestall any confusion with local versus global explanations (input-level versus domain-level reasoning). Both notions specialize uniformly to *every* explanation form we analyze: a cardinally minimal local probabilistic δ -sufficient reason, for example, is the qualifying S of smallest cardinality $|S|$, whereas a subset-minimal one is any qualifying S admitting no qualifying proper subset $S' \subsetneq S$.

5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

The central observation underpinning our framework is that all of the computational problems introduced above — local or global, sufficient or contrastive, probabilistic or non-probabilistic — collapse onto a single template: find a minimum-size subset S with $v(S) \geq \delta$, where the non-probabilistic variants correspond exactly to the choice $\delta := 1$. To make this uniformity explicit, we fix notation for the four value functions. Let v_{suff}^{ℓ} denote the local sufficiency probability of equation 5, namely $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, and let v_{suff}^g denote its global counterpart (equation 6); analogously, let v_{con}^{ℓ} and v_{con}^g denote the local and global contrastive probabilities (equations 7 and 73, respectively). With this notation fixed, the unified task of computing a cardinally minimal local/global sufficient/contrastive δ -reason is stated precisely as follows:

Cardinally Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and $|S|$ is minimal.

For the relaxed objective of computing a subset-minimal, rather than a cardinally minimal, local or global sufficient/contrastive reason, the corresponding task is defined as follows:

Subset Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and for any $S' \subsetneq S$ it holds that $v(S') < \delta$.

Properties of v that govern the complexity. We now isolate the three structural properties of the value function that, as our subsequent analysis establishes, jointly dictate the complexity of computing explanations. The first is *non-decreasing monotonicity*, which guarantees that the marginal contribution $v(S \cup \{i\}) - v(S)$ of any feature is invariably non-negative. Formally:

Definition 2. We say that a value function v maintains non-decreasing monotonicity if for any $S \subseteq \{1, \dots, n\}$ and any $i \in \{1, \dots, n\}$ it holds that: $v(S \cup \{i\}) \geq v(S)$.

The remaining two properties, *supermodularity* and *submodularity*, govern how this marginal contribution $v(S \cup \{i\}) - v(S)$ evolves as the set S grows. In the supermodular case the marginal contribution is itself non-decreasing in S — a feature becomes more valuable the more context has already been fixed — whereas in the dual submodular case it is non-increasing, so that features exhibit diminishing returns. Formally:

Definition 3. Let there be some value function v , some $S \subseteq S' \subseteq \{1, \dots, n\}$, and $i \notin S'$. Then:

1. v maintains supermodularity iff it holds that: $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$.
2. v maintains submodularity iff it holds that: $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$.

6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

In the local non-probabilistic setting, prior work shows that subset-minimal sufficient or contrastive reasons are computable precisely because the value function is monotone — yet this monotonicity holds only in the restricted regime $\delta = 1$, which severely curtails practical flexibility. The natural remedy of passing to probabilistic guarantees for arbitrary δ was shown to backfire: monotonicity collapses in the local probabilistic setting, and computing explanations becomes correspondingly harder (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b). It is exactly this obstruction that our global analysis overcomes.

The global non-probabilistic setting is more extreme still: Bassan et al. (2024) established a stringent *uniqueness* property — there exists exactly *one* subset-minimal explanation — but again only under $\delta = 1$, an assumption especially restrictive globally, where the sufficiency condition must hold for *all* inputs. Indeed, Bassan et al. (2024) shows that this unique minimal subset coincides with the set of all non-strictly-redundant features, which in practice may exhaust the entire feature set. We prove that in the general global *probabilistic* setting — for arbitrary δ — this uniqueness property fails emphatically: the number of subset-minimal global sufficient reasons can be *exponential*.

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Although uniqueness thus survives only in the degenerate case $\delta = 1$, the property that genuinely matters for computation — *monotonicity* — holds for the global value function across *all* values of δ , and for both the sufficient (v_{suff}^g) and the contrastive (v_{con}^g) value functions. This is a striking reversal of the local picture, in which the analogous functions (v_{suff}^l and v_{con}^l) are provably non-monotone:

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^l and v_{suff}^l are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Beyond monotonicity — which holds globally yet fails locally — the global setting exhibits further structure with no local analogue, namely submodularity and supermodularity. Under the standard assumption of feature independence, we prove that the global sufficient value function v_{suff}^g is supermodular, whereas its local counterpart v_{suff}^l fails supermodularity even under the far more restrictive uniform (and hence independent) distribution. Specifically:

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^l is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

For the second family of explanations — global probabilistic *contrastive* reasons — the corresponding value function exhibits the dual property: it is not supermodular but *submodular*. This exact complementarity between the sufficient and contrastive cases again stands in sharp contrast to the local setting, in which the value function is neither submodular nor supermodular.

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

7 COMPUTATIONAL COMPLEXITY RESULTS

7.1 SUBSET MINIMAL EXPLANATIONS

We now analyze the complexity of computing *subset-minimal* explanations (local/global, sufficient/contrastive) across the model classes under study. The decisive property throughout is *mono-*

tonicity. As established in the previous section, monotonicity holds for both global value functions, v_{con}^g and v_{suff}^g , yet fails for both local ones, v_{con}^l and v_{suff}^l . This single distinction is exactly what causes a greedy procedure to converge to a subset-minimal explanation globally while failing to do so locally, and it yields the central finding of this subsection: computing subset-minimal global explanations is tractable (polynomial-time solvable), whereas computing their local counterparts is hard. We begin with the following generalized greedy algorithm:

Algorithm 1 Subset Minimal Explanation Search

Input Value function v , and some $\delta \in [0, 1]$

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1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$ 

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$\triangleright S$ is a (subset minimal?) δ -explanation

Algorithm 1 computes a candidate subset-minimal δ -explanation via a top-down greedy strategy. It initializes S to the full feature set $\{1, \dots, n\}$ and, at each iteration, removes the feature j that inflicts the least damage on the objective — the index j maximizing $v(S \setminus \{j\})$ — continuing as long as some single-feature removal still keeps the value at or above δ (that is, while $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$). Upon termination it returns S .

Whether Algorithm 1 in fact returns a subset-minimal explanation hinges entirely on the *monotonicity* of v . The algorithm halts once removing any single feature from S drives $v(S \setminus \{i\})$ below δ ; monotonicity then upgrades this local stopping condition into the far stronger global guarantee that $v(S \setminus S') < \delta$ for every nonempty S' — which is precisely subset-minimality. Combining this observation with the monotonicity of the global value functions established in the previous sections yields the following proposition:

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^l and v_{suff}^l does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Building on this convergence guarantee, we now establish new complexity results for computing the various forms of subset-minimal explanations within our framework, proceeding model class by model class.

Decision trees. Consider first the most benign, ostensibly “interpretable” regime: f a decision tree and $\mathcal{D}$ an independent distribution. Even here a strict separation emerges. Arenas et al. (2022) proved the intractability result that, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm computes a subset-minimal local δ -sufficient reason for decision trees under independent distributions — indeed, not even under the uniform distribution. We prove that this very same problem becomes tractable in the global regime: a subset-minimal global δ -sufficient reason for decision trees is computable in polynomial time. Formally:

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Extension to other tractable models. The same tractability guarantee for global explanations extends to *orthogonal DNFs* (Crama & Hammer, 2011) — disjunctive normal form formulas whose terms are pairwise mutually exclusive. This extension carries genuine weight, since orthogonal DNFs strictly *generalize* decision trees while retaining the structural properties that render the value function efficiently computable; our guarantee therefore reaches strictly beyond tree-structured models. The full argument appears in Appendix N.

Neural networks, tree ensembles, and other complex models. We now extend these results well beyond decision trees. Whenever $\mathcal{D}$ is an empirical distribution and f admits polynomial-time inference — a mild condition satisfied by essentially every model deployed in practice — computing a

subset-minimal global δ -sufficient or contrastive reason is achievable in polynomial time, irrespective of the model’s internal complexity.

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

This positive result, holding for *any* model under an empirical distribution, sets the stage for a further sharpening of the local–global separation. For sufficiently expressive models, computing subset-minimal *local* explanations remains intractable even when the distribution is empirical. We establish this hardness for both neural networks and tree ensembles, thereby obtaining the following strict complexity separation:

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $PTIME=NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

We now turn from subset-minimal reasons to the more demanding task of computing a *cardinally minimal* δ -sufficient/contrastive reason. In the global setting, the interplay between monotonicity and submodularity/supermodularity yields *provable approximation* guarantees for this task; in the local setting, by contrast, we prove that even bounded approximation is unattainable. This widens the local–global tractability gap from an exact-computation phenomenon into an approximation-theoretic one, and thereby demonstrates that the divide is no artifact of insisting on exact solutions.

A unified greedy approximation algorithm. For a monotone non-decreasing submodular value function, computing a cardinally minimal explanation is an instance of the *submodular set cover* problem (Wolsey, 1982); for a monotone *supermodular* function, it becomes the harder non-submodular variant (Shi et al., 2021). Both are classical objects of study, valued for their strong approximation guarantees (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). The canonical solver for submodular set cover is a greedy algorithm, which we present next (Algorithm 2); it serves as the common engine behind our approximations of both the cardinally minimal sufficient and the cardinally minimal contrastive δ -reason, whose respective guarantees we analyze in turn.

Algorithm 2 Cardinally Minimal Explanation Approximation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$   $\triangleright |S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation

```

Algorithm 2 is the bottom-up dual of Algorithm 1: beginning from the empty set, it greedily augments S , at each step adjoining the feature that maximizes $v(S \cup \{i\})$, until the value first attains the threshold δ .

Cardinally minimal contrastive reasons. We treat global contrastive reasons first, since here both monotonicity and submodularity hold, casting the task exactly as *submodular set cover* and permitting us to invoke the classical guarantee of Wolsey (1982) through Algorithm 2. For integer-valued objectives the algorithm attains a Harmonic-based factor, and in general an $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an *empirical* distribution $\mathcal{D}$ of fixed sample size this ratio collapses to a *constant* — growing only *logarithmically* in the sample size — and hence furnishes a strong guarantee. The *local* setting could scarcely stand in sharper contrast: even for a *single* sample point, no *bounded* approximation exists, marking a categorical gap between the two regimes.

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature*

independence. In contrast, unless $PTIME=NP$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.

We further establish a matching *lower bound* for Theorem 3 in the case of an empirical distribution carrying no independence assumption, thereby certifying that the guarantee is essentially tight (see Appendix M).

Cardinally minimal sufficient reasons. Whereas cardinally minimal global *contrastive* reasons map onto submodular set cover and thereby inherit strong approximations, cardinally minimal global *sufficient* reasons correspond to the supermodular variant, which is inherently harder to approximate — yet still admits guarantees once the function’s curvature is bounded (Shi et al., 2021). Recall the *total curvature* of $v : 2^n \rightarrow \mathbb{R}$, defined as $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. Invoking Shi et al. (2021), we prove that Algorithm 2 attains an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation, so that a provable guarantee persists even in this harder supermodular regime.

As in the contrastive case, under a fixed empirical distribution this approximation is constant. The sufficient case is quantitatively weaker — it carries the additional $\frac{1}{1-k^f}$ factor absent from the tighter $\mathcal{O}(\ln(|D|))$ contrastive bound — yet it remains a genuine constant-factor approximation. And once again the local variant stands in stark opposition: it is inapproximable, admitting no *bounded* approximation even when the dataset collapses to $|D| = 1$.

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{sup}}^g([n])}{\min_{i \in [n]} v_{\text{sup}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $PTIME=NP$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Taken together, these results reinforce and extend Subsection 7.1: having shown that subset-minimal global explanations are tractable where their local counterparts are hard, we have now shown that even the harder cardinally minimal global explanations are efficiently approximable, whereas their local counterparts remain inapproximable. The local–global divide thus proves remarkably robust, persisting across both minimality notions and across both the exact and the approximate regimes.

8 LIMITATIONS AND FUTURE WORK

Our most fundamental findings — the structural properties of the value functions — hold in full generality, and we instantiate them to obtain concrete complexity results for specific model classes (e.g., neural networks), distributional assumptions (e.g., empirical distributions), and explanation definitions within our framework. Settings beyond those instantiated here remain open, and we regard this breadth of unexplored terrain not as a limitation but as a measure of the framework’s reach: the general structural results, together with the new tractability and intractability separations they yield, constitute a principled foundation on which a far wider range of explainability scenarios can be analyzed with the very same machinery.

Our study opens several concrete theoretical directions:

1. First, the tractable global-explainability results established here may admit further tightening for structurally simple model classes such as decision trees, linear models, and other inherently tractable models. Although local explanations for these models are frequently intractable (Arenas et al., 2022; Subercaseaux et al., 2025), the strong structural properties of the global explanation forms we study already yield markedly improved guarantees, and we conjecture that these can be sharpened further; tightening the approximation factors under empirical distributions is an especially promising avenue.
2. Second, when the relevant expectations cannot be computed *exactly*, it is natural to ask whether they can instead be approximated — for instance, via Fully Polynomial Randomized Approximation Schemes (FPRAS) for model classes such as DNF formulas (Meel et al., 2019) (and hence tree ensembles), or via Monte Carlo estimators of the kind employed in (Subercaseaux et al., 2025). The central challenge is to characterize the tolerance within which the minimality guarantees provably survive approximation error.

-
3. Finally, we anticipate that alternative notions of feature importance — for example, loss measures such as KL divergence (Conmy et al., 2023) — exhibit the same local-to-global emergence of monotonicity, submodularity, or supermodularity, which would extend efficient feature selection to those metrics as well.

9 CONCLUSION

We have introduced a unified framework for analyzing diverse explanation types and used it to expose a stark structural contrast between local and global sufficient and contrastive reasons. The *local* variants possess none of the properties central to combinatorial optimization — neither monotonicity, submodularity, nor supermodularity — whereas we prove that their *global* counterparts possess exactly these properties: (i) *monotonicity*, (ii) *submodularity* for contrastive reasons, and (iii) *supermodularity* for sufficient reasons. These structural results form the foundation for a series of complexity separations establishing that global explanations with provable guarantees are efficiently computable — even for complex model classes such as neural networks — exactly where their local counterparts remain NP-hard even in drastically simplified settings. By uncovering these foundational properties and charting the precise frontier between tractability and intractability, our work supplies both the conceptual framework and the technical machinery for a broad program of future research on the computation of formal explanations.

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Appendix

The appendix contains formalizations and proofs that were mentioned throughout the paper:

Appendix A contains extended related work on Formal XAI.

Appendix B contains the formalizations of the models and distributions used in this work.

Appendix C contains the proof of Proposition 1.

Appendix D contains the proof of Proposition 2.

Appendix E contains the proof of Proposition 3.

Appendix F contains the proof of Proposition 4.

Appendix G contains the proof of Proposition 5.

Appendix H contains the proof of Theorem 1.

Appendix I contains the proof of Proposition 6.

Appendix J contains the proof of Theorem 2.

Appendix K contains the proof of Theorem 3.

Appendix L contains the proof of Theorem 4.

Appendix O contains an LLM usage disclosure.

A EXTENDED RELATED WORK ON FORMAL XAI

Our work belongs to the research program of *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), which studies explanations endowed with provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). A central theme of this program is the computational complexity of generating such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Since formal explanations are typically intractable to compute for expressive models such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), much of the literature has retreated to structurally simpler model classes (Marques-Silva & Ignatiev, 2023), among them decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026). Against this backdrop, our global tractability results are notable precisely because they reach the most expressive models — neural networks and tree ensembles — that local analyses have been unable to accommodate.

Complementing these complexity-theoretic investigations, a substantial body of work computes formal explanations in practice through automated reasoning: MaxSAT and MILP solvers for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), and neural-network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) for deriving certified explanations of neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). Further approaches mitigate the inherent computational burden via abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxations such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), or training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Finally, a closely related line of work within formal XAI applies combinatorial and submodular optimization techniques to the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), chiefly for decision-tree classifiers. As that literature observes — and as we confirm here — the value function underlying these local variants is not submodular, so that submodular-inspired greedy methods operate there merely as heuristics, devoid of formal guarantees, however high the quality of the explanations they often yield in practice. Our work instead forges a rigorous connection between combinatorial and submodular optimization objectives and the *global* variants of sufficient and contrastive reasons: because the requisite structure provably holds globally, this connection delivers genuine formal guarantees and thereby opens the door to a broad class of principled algorithmic implementations.

B MODEL AND DISTRIBUTION FORMALIZATIONS

This section provides the formal definitions of the models and distributions referenced throughout the paper: Subsection B.1 defines the model families, while Subsection B.2 formalizes the distribution classes.

B.1 MODEL FORMALIZATIONS

This subsection formalizes the three base-model types analyzed throughout the paper: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations. These formalizations fix the objects to which our model-specific complexity results apply.

Decision Trees. We define a decision tree (DT) as a directed acyclic graph representing a function $f : \mathbb{F} \rightarrow [c]$ over $c \in \mathbb{N}$ classes, whose structure encodes the function as follows: (i) each internal node v is labeled with a distinct binary input feature from $\{1, \dots, n\}$; (ii) each internal node v has at most k outgoing edges, one per value in $[k]$ that may be assigned to v ; (iii) along any root-to-leaf path α , every variable appears at most once; (iv) each leaf is labeled with a class in $[c]$. Consequently, every input assignment $\mathbf{x} \in \mathbb{F}$ traces a unique root-to-leaf path α , and the output $f(\mathbf{x})$ is the class label $i \in [c]$ of the terminal leaf. The size $|f|$ of the DT is the total number of edges in the graph. To afford modeling flexibility, distinct paths α and α' may test the input variables $\{1, \dots, n\}$ in differing orders, subject only to the constraint that no variable is repeated along any single path.

Neural Networks. We state our *hardness* proofs for neural networks with ReLU activations, whereas our *tractability* results (polynomial-time algorithms) require only the standard assumption of polynomial-time inference. Every tractable-versus-intractable separation we prove therefore transfers to *any* neural architecture at least as expressive as a standard feed-forward ReLU network — a class that subsumes the models most widely deployed in practice, including ResNets, CNNs, Transformers, and diffusion models. Since any ReLU network can be rendered fully connected by assigning zero weights and biases to the absent connections, we follow standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolphi et al., 2025) and assume full connectivity without loss of generality; that is, our analysis applies to multi-layer perceptrons (MLPs). Formally, an MLP f comprises $t - 1$ *hidden layers* $(g^{(j)})$ for $j = 1, \dots, t - 1$ together with one output layer $g^{(t)}$, each layer being defined recursively by:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where $W^{(j)}$, $b^{(j)}$, and $\sigma^{(j)}$ denote, respectively, the weight matrix, bias vector, and activation function of the j -th layer. The model is accordingly parameterized by t weight matrices $(W^{(1)}, \dots, W^{(t)})$, t bias vectors $(b^{(1)}, \dots, b^{(t)})$, and t activation functions $(\sigma^{(1)}, \dots, \sigma^{(t)})$.

The input layer is $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$, the binary input vector. The network’s dimensions are fixed by a sequence of positive integers $d_0, \dots, d_t$, and the (training-learned) parameters satisfy $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$ and $b^{(j)} \in \mathbb{Q}^{d_j}$. Since the model is a binary classifier over n features, we set $d_0 = n$ and $d_t = 1$. The hidden layers employ the ReLU activation $\text{ReLU}(x) = \max(0, x)$. While training typically relies on a sigmoid output activation, for the purpose of post-hoc interpretation we take the output layer to apply the threshold-based step function $\text{step}(\mathbf{z}) = 1$ if $\mathbf{z} \geq 0$ and $\text{step}(\mathbf{z}) = 0$ otherwise.

Tree Ensembles. Since our goal is post-hoc explanation, we characterize ensembles by their *inference* behavior rather than by their training procedure. Our analysis targets the two dominant inference schemes: *majority voting* (e.g., random forests) and *weighted voting* (e.g., XGBoost). As with neural networks, however, our *tractability* results — the polynomial-time algorithms — require only polynomial-time inference, and hence extend to an even broader range of ensemble configurations.

Majority Voting Inference. Under *majority voting*, the prediction $f(\mathbf{x})$ is the class $j \in [c]$ that garners the plurality of votes among the individual tree predictions $f_i(\mathbf{x})$ over all $i \in [k]$:

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

where $\text{majority}(S)$ returns the most frequent label in the multiset S , ties being resolved by a fixed rule (e.g., lexicographic order or a predefined priority).

Weighted Voting Inference. Under *weighted voting*, each ensemble member carries a weight $\phi_i \in \mathbb{Q}$ encoding its relative importance, and the predicted class is the one accumulating the greatest total weight. Formally, for any $\mathbf{x} \in \mathbb{F}$:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where $\mathbb{I}$ denotes the indicator function.

B.2 DISTRIBUTION FORMALIZATIONS

This subsection formalizes the three distribution classes introduced in the main paper.

Empirical Distributions. An empirical distribution $\mathcal{D}$ over the input features is induced by a dataset $\mathbf{D}$ of inputs $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$, and assigns each $\mathbf{x} \in \mathbb{F}$ a probability equal to its relative frequency in $\mathbf{D}$:

$$\mathbf{Pr}(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

Independent Distributions. Given a per-feature marginal $p(\mathbf{x}_i) \in [0, 1]$ for each input feature, the distribution $\mathcal{D}$ is *independent* precisely when the joint probability factorizes as $\mathbf{Pr}(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$.

When $p(\mathbf{x}_i) = p(\mathbf{x}_j)$ for all $i, j \in [n]$, this specializes to the *uniform distribution*, itself a special case of an independent distribution.

General Distributions. Many of our proofs — especially those establishing the fundamental properties of the value functions — hold under fully general distributional assumptions, and this generality extends their applicability well beyond any single modeling choice. Although empirical distributions (using the training set as a proxy) are the prevailing choice in XAI, our framework accommodates alternative distribution estimators equally well, among them k -NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for tabular dependencies (Größer & Okhrin, 2022), and expressive conditional generative models such as CTGAN (Xu et al., 2019) and conditional diffusion models (Huang et al., 2022) — estimators widely used in the counterfactual-explanation and algorithmic-recourse literature (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024) to which contrastive explanations are closely related. When the value function is derived neither from a structural property of the model (e.g., leaf enumeration in decision trees) nor from an empirical distribution, it can instead be estimated by Monte Carlo sampling (Hastings, 1970), following approaches akin to (Subercaseaux et al., 2025; Lopardo et al., 2023).

B.3 SUBSET VS. CARDINAL MINIMALITY

This subsection examines the distinction between cardinal and subset minimality in greater detail. Cardinal minimality is the substantially stronger guarantee, corresponding to globally minimal explanation size, whereas subset minimality secures only a local form of minimality.

That cardinal minimality implies subset minimality is immediate: if $S \subseteq [n]$ is cardinally minimal, then no set S' with $|S'| < |S|$ is an explanation, so in particular no strict subset $S' \subsetneq S$ is an explanation, whence S is subset-minimal.

The converse, however, fails. Consider the function:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

together with the assignment $\mathbf{x} := (1, 1, 1)$, so that $f(1, 1, 1) = 1$. Fixing $\mathbf{x}_1 = 1$ yields a cardinally minimal (and hence subset-minimal) sufficient reason, since the prediction remains 1 irrespective

of $\mathbf{x}_2, \mathbf{x}_3$. Fixing both $\mathbf{x}_2 = 1$ and $\mathbf{x}_3 = 1$ likewise yields a subset-minimal explanation — yet not a cardinally minimal one, as it has strictly larger cardinality. This confirms that every cardinally minimal explanation is subset-minimal, but not conversely.

C PROOF OF PROPOSITION 1

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Proof. It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input $\mathbf{x}$ Bassan et al. (2024), a phenomenon exhibited by functions of the form below. We prove that this same function admits an exponential number of *global* and *probabilistic* sufficient reasons, thereby transferring the exponential blow-up to our setting. The function is a special case of the broader class of *threshold Boolean functions* of Wegener (2005): for $n = 2k + 1$ with $k \in \mathbb{N}$, it is defined by:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\mathbf{x}$ is drawn from a uniform distribution $\mathcal{D}$. The function f is *symmetric* in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends solely on the number of 1's (equivalently, 0's) in the input and is invariant under every permutation of the input bits. Moreover, the two extreme inputs — the all-zeros vector $(0, \dots, 0)$ and the all-ones vector $(1, \dots, 1)$ — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons, and it is this that drives the count.

Because f is symmetric, the equality $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$ holds for any two subsets $S_1, S_2 \subseteq [n]$ of equal size. Furthermore, one verifies directly that for any subset $S \subseteq [n]$ of size $k + 1$:

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set δ to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and set $k := \lfloor \frac{n}{2} \rfloor$. Each such subset is *subset-minimal*, since every subset of size at most $\lfloor \frac{n}{2} \rfloor - 1$ fails to be a sufficient reason for $\langle f, \mathbf{x} \rangle$ at the threshold δ . The counting argument of (Bassan et al., 2024) then applies verbatim, yielding exactly $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subset-minimal local sufficient reasons for $\langle f, \mathbf{x} \rangle$. By Stirling's approximation:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This establishes the claimed $\Theta(\frac{2^n}{\sqrt{n}})$ bound on the number of subset-minimal global δ -sufficient reasons, thereby completing the proof. $\square$

D PROOF OF PROPOSITION 2

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Proof. We establish monotonicity for both families of global explanations — sufficient and contrastive. We begin with global sufficiency, first treating the illustrative Boolean case to build intuition, then generalizing successively to functions with discrete multi-valued input domains and multiple output classes, and finally to continuous input domains, thereby covering every well-defined classification function. From this we then derive the corresponding guarantee for global contrastive explanations. Crucially, none of these proofs invokes feature independence: monotonicity holds in the strongest possible sense, for *any* underlying distribution.

We close by exhibiting a simple example that demonstrates the failure of monotonicity for the local probabilistic sufficient and contrastive value functions, thereby confirming that the property is genuinely a global phenomenon.

Lemma 1. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for Boolean functions, under any data distribution.*

Proof. Given an arbitrary set $S \in [n]$, and a feature $i \notin S$, we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$ such that $\mathbf{x}_S = \mathbf{x}'_S$ and $f(\mathbf{x}) = 1 - f(\mathbf{x}')$, it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') \mid \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

For notational economy, we occasionally suppress the $\sim \mathcal{D}$ subscript in the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, and write f^+ and f^- for the events $f(\mathbf{z}) = 1$ and $f(\mathbf{z}) = 0$, respectively.

We first carry out a technical simplification of the probability term, which streamlines the subsequent monotonicity argument.

Simplifying the probability term. Let $\mathcal{D}_S$ denote the distribution restricted to S . Once a point $\mathbf{x} \sim \mathcal{D}$ and a set S are fixed, all points $\mathbf{z} \sim \mathcal{D}$ with $\mathbf{z}_S = \mathbf{x}_S$ and $f(\mathbf{z}) = b$, for $b \in \{0, 1\}$, share one and the same local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S)$. Marginalizing over $x_{\bar{S}}$ across all $\mathbf{x} \sim \mathcal{D}$ that agree on x_S and yield the same output b gives $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$, whence:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S). \quad (21)$$

Thus, for $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$, we get that:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S))^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2], \end{aligned} \quad (22)$$

From which it follows that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\
&\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2]
\end{aligned} \tag{23}$$

Proving monotonicity. Building on the preceding simplification, we now establish the monotonicity claim, first introducing some additional notation. Let $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$ and $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$, and, analogously, $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$ and $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$. Then, the following expectations $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ and $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$ can be simplified accordingly¹:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{24}$$

As a result, we obtain the following:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{25}$$

Let $\Delta_t(S, f)$ denote $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$. Without loss of generality, we assume that both P_1 and P_0 are positive. Otherwise, it can be verified that $\Delta_t(S, f) = 0$. Then, we get:

¹Note that the notations P_1 , P_0 , g_1^+ , and g_0^+ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input $\mathbf{x}_S$. Formally, they should be written as $P_1(\mathbf{x}_S)$, $P_0(\mathbf{x}_S)$, and so on. However, for readability, we omit the explicit dependence on $\mathbf{x}_S$.

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\
&\geq 0.
\end{aligned} \tag{26}$$

Here we have invoked the identities $P_1 - P_1^2 = P_1(1 - P_1) = P_1P_0$ and, symmetrically, $P_0 - P_0^2 = P_0(1 - P_0) = P_0P_1$. Together with the preceding derivation, these establish that $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, thereby completing the proof. $\square$

Lemma 2. *The global sufficient value function v_{uff}^g is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

Proof. We now lift the Boolean argument to functions whose input domains and output ranges are each finite sets of values, letting $\mathcal{K}$ denote the set of output classes. Equation 22 then generalizes to:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \\
&\sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right]
\end{aligned} \tag{27}$$

Conditioning on $\mathbf{z}_S = \mathbf{x}_S$ and letting feature t range over its r possible values $\{v_1, \dots, v_r\}$, equation 23 generalizes to:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{k=1}^r \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_t = v_k \mid \mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right].
\end{aligned} \tag{28}$$

Let P_k denote $\Pr(z_t = v_k \mid \mathbf{x}_S)$, Let g_k^j denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$, that is, z_t takes the value v_k and the output class is j . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \tag{29}$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{30}$$

Therefore $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, thereby completing the proof for the discrete multi-valued setting. $\square$

Lemma 3. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

Proof. We extend monotonicity to functions with continuous feature domains. One route partitions each continuous feature’s domain into two disjoint subdomains and adapts the proof of lemma 2 through a change of notation alone; since both summation and integration are linear, the core reasoning carries over intact from the discrete to the continuous case.

For completeness, we instead present a direct proof. Assume all features lie in continuous domains, and let $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$ denote the joint probability density, $p(x_i)$ the marginal density of x_i , and $\mathcal{X}_i$ the domain of feature i .

We define the marginal density by

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \tag{31}$$

where $\mathcal{X}_{\bar{S}}$ is the domain of the features in $\bar{S}$, and the conditional density by:

$$p(x_t \mid \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \tag{32}$$

The local probability of S given $\mathbf{x}$ is then

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \tag{33}$$

which decomposes, via an additional feature $t \notin S$, as:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \tag{34}$$

Equation 27 accordingly becomes

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S,
\end{aligned} \tag{35}$$

while equation 28 becomes

$$\begin{aligned} & \mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For brevity, write $Y_j(t)$ for $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$ and $p_{t|S}$ for $p(z_t \mid \mathbf{x}_S)$; both are implicitly tied to a fixed $\mathbf{x}_S$ ². Monotonicity then follows from:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

□

Proposition 2. While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.

Proof. Combining lemmas 1 to 3, we conclude that the global sufficient value function v_{suff}^g is monotonic non-decreasing for every well-defined classification function — Boolean, discrete, continuous, or any hybrid thereof — under any data distribution.

$$v_{con}^g(S) = 1 - v_{suff}^g(\bar{S}) \quad (38)$$

Hence v_{con}^g is monotonic non-increasing as features are added to $\bar{S}$ or, equivalently, monotonic non-decreasing as features are removed from $\bar{S}$ and adjoined to S , thereby completing the proof.

□

E PROOF OF PROPOSITION 3

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.

Proof. This section establishes supermodularity for both global sufficient and contrastive explanations. As with monotonicity, we proceed from Boolean functions to multi-valued discrete and then continuous input domains, ultimately covering every well-defined classification function. In contrast

²Changing $\mathbf{x}_S$ alters their values; we suppress $\mathbf{x}_S$ in the notation for brevity.

to monotonicity, however, these results require the assumption of feature independence, and we substantiate this requirement with a counterexample demonstrating that supermodularity does indeed fail once independence is relaxed.

Lemma 4. *The global sufficient value function v_{uff}^g is supermodular for Boolean functions, under the assumption of feature independence.*

Proof. Concretely, we prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

for $S \subseteq S'$ and $t \notin S'$ — that is, that the marginal gain $\Delta_t(S, f)$ is itself monotone in S .

The case $S' = S$ is immediate, so it suffices to treat $S' = S \cup \{i\}$ with $i \notin S$ and $i \neq t$. Under feature independence, the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$ decomposes into:

$$\begin{aligned} \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) &= \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ &\Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ &\Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ &\Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Write P_1, P_0, Q_1 , and Q_0 for $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$, and $\Pr(z_i = 0)$, respectively, and let $g_{11}^+, g_{10}^+, g_{01}^+$, and g_{00}^+ stand for $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$, and $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$, respectively.

We further decompose equation 26 into:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

This shows that $\Delta_t(S, f)$ is monotone; that is, $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$. $\square$

Lemma 5. *The global sufficient value function v_{uff}^g is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

Proof. We extend the results of the previous section to multi-valued discrete functions, once more under the assumption of feature independence. We use P_k to denote $\Pr(z_t = v_k)$, and $Q_{k'}$ to denote $\Pr(z_i = v_{k'})$. Moreover, we let $g_{k'k}^j$ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v_{k'}, z_t = v_k)$. We decompose equation 30 into

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} P_k P_l \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} P_k P_l \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right]^2. \end{aligned} \quad (44)$$

Likewise, we have

$$\begin{aligned}
\Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v'_k}} \mathbf{Pr}(\mathbf{x}_S) \cdot \mathbf{Pr}(z_i = v'_k) \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right].
\end{aligned} \tag{45}$$

Therefore, we get

$$\begin{aligned}
&\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{46}$$

□

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{diff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{diff}^g is supermodular.*

Proof. We can generalize the supermodularity results established in Lemma 5 to the continuous case, provided that $\mathcal{D}_p$ satisfies feature independence. One approach partitions the domain of the continuous features into two disjoint subdomains and then adapts the proofs in Lemma 5 through a corresponding revision of notation.

Alternatively, we present a direct proof below. Under this assumption, the marginal and conditional densities simplify as follows:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t | \mathbf{x}_S) = p(x_t). \tag{47}$$

Consequently, the local probability of S conditioned on $\mathbf{x}$ takes the form:

$$\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \tag{48}$$

We let $Y_j(i, t)$ denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$, where $i \notin S$, $t \notin S$ and $i \neq t$. Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
&\Delta_t(S, f) \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\ &= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}. \end{aligned}$$

Let $S' = S \cup \{i\}$, then we have:

$$\begin{aligned} & \Delta_t(S', f) \\ &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \quad (50) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S, \end{aligned}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\ &= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}. \end{aligned}$$

Let $\alpha(i, t, t')$ denote $Y_j(i, t) - Y_j(i, t')$ and $\beta(i', t, t')$ denote $Y_j(i', t) - Y_j(i', t')$, we can infer that

$$\begin{aligned} & \Delta_t(S', f) - \Delta_t(S, f) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[\sum_{j \in \mathcal{K}} \frac{1}{4} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (51)$$

Consequently, the global sufficient value function v_{suff}^g retains supermodularity for every valid classification function, irrespective of whether its features are Boolean, discrete, continuous, or any hybrid thereof.

We now demonstrate that the local probabilistic sufficient value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution over the three variables $\{x_1, x_2, x_3\}$ and define the Boolean function:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \quad (52)$$

Consider the input $\mathbf{x} = (1, 1, 1)$, which is classified as 1. For the local probabilistic sufficient value function v_{suff}^l , the following values hold:

$$\begin{aligned} v_{\text{suff}}^l(\emptyset) &= \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4}, \\ v_{\text{suff}}^l(\{1, 2\}) &= \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1. \end{aligned}$$

Since $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$, the function v_{suff}^l is neither supermodular nor submodular. □

F PROOF OF PROPOSITION 4

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

Proof. Once the global sufficient setting has been shown to be supermodular, it follows directly that the global contrastive setting is submodular. Let $S \subseteq [n]$ be the set of features allowed to change, and suppose $S \subseteq S'$ with $i \notin S'$. To compare the marginal contributions $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$ and $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$, we equivalently compare $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$ and $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$. Since $S \subseteq S'$, it follows that $\bar{S}' \subseteq \bar{S}$, and therefore we obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus, the global contrastive value function v_{con}^g is submodular, thereby concluding this proof.

We now demonstrate that the local probabilistic contrastive value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution over the three variables $\{x_1, x_2, x_3\}$ and define the Boolean function:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input $\mathbf{x} = (1, 1, 1)$, which is classified as 1. For the local probabilistic contrastive value function v_{con}^l , we obtain:

$$\begin{aligned} v_{\text{con}}^l(\emptyset) &= 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2}, \\ v_{\text{con}}^l(\{1, 2\}) &= \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}. \end{aligned}$$

Since $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$, the function v_{con}^l is neither submodular nor supermodular. $\square$

In the remainder of this section, we show that the feature independence assumption is not merely sufficient but in fact necessary for supermodularity. We establish this through the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

For this Boolean function, we have $f(00) = 0$, $f(01) = 0$, $f(10) = 1$, and $f(11) = 1$. Now fix an arbitrary distribution $\mathcal{D}$ over x_1 and x_2 , and recall that the global sufficient value function is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \mathbf{Pr}(\mathbf{x}) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, establishing that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

for $S \subseteq S'$ and $t \notin S'$ would certify that the global sufficient value function is supermodular. When $S = \emptyset$, we have:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) \\
& = (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2
\end{aligned}$$

When $S = \{2\}$, the following condition holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1)
\end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\
& = \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)}
\end{aligned}$$

By contrast, when $S = \{1\}$, the following holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1)
\end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1
\end{aligned}$$

Finally, when $S = \{1, 2\}$, the following condition holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1)
\end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned}
& \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0)} + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1)} + \\
& \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 0)}{\mathbf{Pr}(x_1 = 1, x_2 = 0)} + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 1)}{\mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= 1
\end{aligned}$$

Let $t = 2$. Evidently, $\Delta_t(\{1\}, f) = 0$. Next, we analyze $\Delta_t(\emptyset, f)$. Under the assumption of feature independence, we observe that:

$$\begin{aligned}
& (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2,
\end{aligned}$$

Additionally, another condition that holds is:

$$\begin{aligned}
& \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 0)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 1)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2.
\end{aligned}$$

This yields $\Delta_t(\emptyset, f) = 0$; that is, $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$. Absent the assumption of feature independence, however, we can no longer guarantee $\Delta_t(\emptyset, f) = 0$. For instance, let:

$$\mathbf{Pr}(x_1 = 0, x_2 = 0) = P_{00}, \mathbf{Pr}(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\mathbf{Pr}(x_1 = 1, x_2 = 0) = P_{10}, \mathbf{Pr}(x_1 = 1, x_2 = 1) = P_{11}$$

Then, simplifying $\Delta_t(\emptyset, f)$ yields:

$$\Delta_t(\emptyset, f) = \left[\frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[(P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

Setting $P_{00} = 0.4$, $P_{01} = 0.2$, $P_{10} = 0.1$, and $P_{11} = 0.3$ then gives:

$$\Delta_t(\emptyset, f) = \left[\frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[(0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This implies that $\Delta_t(\emptyset, f) > \Delta_t(1, f)$, showing that — in contrast to the monotonicity setting, where it was dispensable — feature independence is a necessary condition for supermodularity.

G PROOF OF PROPOSITION 5

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Proof. This proposition follows directly from our proof that both v_{suff}^g and v_{con}^g are monotone non-decreasing (as established in Proposition 2), together with related conclusions drawn in prior work. In the remainder of this section, we make explicit how this monotonicity property yields the stated corollary.

The convergence of Algorithm 1 to a subset-minimal δ global sufficient or contrastive reason with respect to the value functions v_{con}^g and v_{suff}^g is an immediate consequence of monotonicity: prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) establishes that this class of greedy algorithm yields a subset-minimal explanation whenever the underlying value function is monotone non-decreasing. The algorithm halts once, for every $i \in S$, the value $v(S \setminus \{i\})$ drops below δ , so that although $v(S) \geq \delta$ is maintained as an invariant, removing any single element forces the condition to fail. By monotonicity, this in turn implies that no proper subset of S satisfies the condition, guaranteeing that S is subset-minimal.

We now explain why Algorithm 1 fails to converge to a subset-minimal solution in the local setting. It is well established that the local probabilistic sufficient value function (and, likewise, the local probabilistic contrastive value function) is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). As a simple illustration, fix the uniform distribution over the two binary variables $\{x_1, x_2\}$ and define the Boolean function:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input $\mathbf{x} = (0, 1)$, which is classified as 1; then:

$$v_{\text{suff}}^\ell(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^\ell(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^\ell(\{2\}) = 1, \quad v_{\text{suff}}^\ell(\{1, 2\}) = 1. \quad (60)$$

Since $v_{\text{suff}}^\ell(\emptyset) > v_{\text{suff}}^\ell(\{1\}) < v_{\text{suff}}^\ell(\{1, 2\})$, the function v_{suff}^ℓ is not monotone. Likewise, one computes:

$$v_{\text{con}}^\ell(\emptyset) = 1, \quad v_{\text{con}}^\ell(\{1\}) = 1, \quad v_{\text{con}}^\ell(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^\ell(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For v_{con}^ℓ , the set S denotes the features that are permitted to vary.) Since $v_{\text{con}}^\ell(\emptyset) > v_{\text{con}}^\ell(\{2\}) < v_{\text{con}}^\ell(\{1, 2\})$, the function v_{con}^ℓ is likewise non-monotone.

□

H PROOF OF THEOREM 1

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Proof. We begin by observing that for decision trees, under the assumption that $\mathcal{D}$ is an independent distribution, the value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

can be evaluated in polynomial time via the following procedure. Exploiting the tractability of decision trees, we iterate over all pairs of leaf nodes, each corresponding to partial assignments $\mathbf{x}_S$ and $\mathbf{z}_S$. For every such pair, whenever both leaf nodes yield the same prediction under f , we compute the corresponding term in the expected value $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ by multiplying the respective feature-wise probabilities over the features shared by the two vectors. Under the feature independence assumption these probabilities decompose, so the full expectation is recovered by summing the contributions of all matching leaf pairs.

Since every probability involved is supplied as part of the input and each step performs only polynomial-time operations, the entire procedure runs in polynomial time. This establishes that every step of Algorithm 1 is efficient, and hence that the algorithm as a whole runs in polynomial time. Combining this with Lemma 2, which proves that v_{suff}^g is monotone non-decreasing, we conclude that Algorithm 1 converges to a subset-minimal explanation: the algorithm halts once, for every $i \in S$, the value $v(S \setminus \{i\})$ falls below δ , while $v(S) \geq \delta$ is preserved throughout, and monotonicity then guarantees that no strict subset of S satisfies the condition, ensuring that S is indeed subset-minimal.

For the remaining part of the claim, the result follows from (Arenas et al., 2022), which established that, assuming $P \neq NP$, no polynomial-time algorithm computes a subset-minimal δ -sufficient reason for decision trees. Although this was proven specifically under the uniform distribution, the hardness extends immediately to independent distributions, which subsume that case. Together, these results yield the complexity separation between the local and global variants, thereby concluding our proof.

□

I PROOF OF PROPOSITION 6

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Proof. The evaluation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

can be performed in polynomial time whenever $\mathcal{D}$ is drawn from the class of empirical distributions. This is achieved by iterating over pairs of instances $\mathbf{x}, \mathbf{z}$ within the dataset and running inference through $f(\mathbf{x})$ and $f(\mathbf{z})$ to accumulate the expected value at each step. It follows that deciding whether the probability function exceeds or falls below a given threshold δ is likewise polynomial-time. Furthermore, leveraging the non-decreasing monotonicity of the global probability function (Proposition D), we may deploy Algorithm 1 for either contrastive-reason search or sufficient-reason search. Executing a linear number of polynomial-time queries then yields a subset-minimal explanation — whether a subset-minimal contrastive reason or a subset-minimal sufficient reason — in polynomial time.

□

J PROOF OF THEOREM 2

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $P = NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

Proof. The first part of the proof is an immediate consequence of Proposition 6, which established that the claim holds for any model, since both $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ and $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ are computable in polynomial time from the empirical distribution. For the second part, we prove two lemmas: one establishing the intractability of the sufficient case, the other that of the contrastive case.

Lemma 6. *Unless $P = NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We first prove the claim for neural networks and then extend the result to tree ensembles. We establish it by showing that a polynomial-time algorithm for computing a subset-minimal local

δ -sufficient reason for neural networks would enable us to solve the classic NP-complete *CNF-SAT* problem in polynomial time. The CNF-SAT problem is defined as follows:

CNF-SAT:

Input: A formula in conjunctive normal form (CNF): ϕ .

Output: *Yes*, if there exists an assignment to the n literals of ϕ such that ϕ is evaluated to True, and *No* otherwise

Our proof will also rely on the following lemma (whose proof is provided in Barceló et al. (2020)):

Lemma 7. *Any boolean circuit ϕ can be encoded into an equivalent MLP over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We in fact establish hardness for a simpler, more restricted case, which then implies hardness for the general setting. Here we assume that the empirical distribution $\mathcal{D}$ consists of a single element, which for simplicity we take to be the zero vector $\mathbf{0}_n$. Because the sufficiency of S in this scenario depends only on the local instance $\mathbf{x}$ and the single instance $\mathbf{0}_n$ in the empirical distribution, the nature of the input format (discrete, continuous, and so on) does not affect the outcome; the hardness result therefore holds uniformly across all such settings.

Analogously to $\mathbf{0}_n$, we define $\mathbf{1}_n$ as the n -dimensional vector consisting entirely of ones. Given an input ϕ , we first assign $\mathbf{1}_n$ to ϕ , thereby setting all variables to True. If ϕ evaluates to True under this assignment, a satisfying truth assignment exists; we may therefore assume $\phi(\mathbf{1}_n) = 0$. We now introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

ϕ' , while no longer a CNF, can still be transformed into an MLP f using Lemma 8, ensuring that f behaves equivalently to ϕ' over the domain $\{0, 1\}^n$. Given our assumption that computing a subset-minimal local δ -sufficient reason is feasible in polynomial time, we can determine one for the instance $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$, $\delta := 1$, noting that the empirical distribution $\mathcal{D}$ is simply defined over the single data point $\mathbf{0}_n$.

We now assert that the subset-minimal δ -sufficient reason generated for $\langle f, \mathbf{x} \rangle$ encompasses the entire input space, i.e., $S = \{1, \dots, n\}$, if and only if $\langle \phi \rangle \notin \text{CNF-SAT}$.

Let us assume that $S = \{1, \dots, n\}$. Since S is a δ -sufficient reason for $\langle f, \mathbf{x} \rangle$, this simply means that setting the complementary set to any value maintains the prediction. Since the complementary set is $\emptyset$ in this case, this trivially holds. The fact that S is subset-minimal means that any other subset $S' \subseteq S$ satisfies $v(S') < \delta = 1$. Since the probability function $\Pr_{\mathbf{x} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$ is determined by a single point (the distribution contains only the point $\mathbf{0}_n$), the probability function can only take values of 1 or 0. Hence, we also know that $v(S') = 0$. This tells us that, aside from the subset $S = \{1, \dots, n\}$, for any subset $S' \subseteq S$, fixing the features in S' to 1 and the rest to 0 does not result in a classification outcome of 1. Since the ϕ_2 component within ϕ' is True only if all features are assigned 1, this directly implies that ϕ is assigned False for any of these inputs. Since we already know that ϕ does not return a True answer for the vector assignment $\mathbf{1}_n$ (as verified at the beginning), and now we have established that the same holds for all other input vectors, we conclude that $\phi \notin \text{CNF-SAT}$.

Now, suppose that S is a subset that is strictly contained within $\{1, \dots, n\}$. Given that S is sufficient under our definitions of the distribution $\mathcal{D}$ with $\delta = 1$, we can apply the same reasoning as before to conclude that $v(S) = 1$. This implies that setting the features in S to 1 while setting the remaining features to 0 ensures that the function f evaluates to 1. Since this assignment is necessarily not a vector consisting entirely of ones, it follows that the ϕ_2 component within ϕ' must be False. Consequently, the ϕ component must be True, which implies that $\langle \phi \rangle \in \text{CNF-SAT}$. This completes the proof.

We have established that the following claim holds for neural networks. However, extending the proof to tree ensembles requires a minor and straightforward adaptation. To do so, we will utilize the following lemma, which has been noted in several previous works (Ordyniak et al., 2024; Audemard et al., 2022b):

Lemma 8. Any CNF or DNF ϕ can be encoded into an equivalent random forest classifier over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.

Proof. We observe that we can apply the same process used in our proof for neural networks, where we encoded ϕ' into an equivalent neural network. However, ϕ' is no longer a valid CNF due to our construction (though encoding it into an MLP was not an issue, as any Boolean circuit can be transformed into an MLP). Nevertheless, since ϕ' consists of a conjunction of only two terms, we can easily represent it as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots (c_m \vee \phi_2) \quad (66)$$

Where each c_i is a disjunction of a few terms. Consequently, ϕ' is a valid CNF, allowing us to transform it into an equivalent random forest classifier. The reduction we outlined for MLPs applies directly to these models as well, thereby completing the proof for both model families.

□

Lemma 9. Unless $PTIME = NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.

Proof. We will present a proof analogous to the one in Lemma 6. Specifically, we will once again utilize the classical NP-hard CNF-SAT problem defined in Lemma 6. In particular, given a Boolean formula ϕ , we will demonstrate that determining a subset-minimal contrastive reason — whether for a neural network or a tree ensemble — allows us to decide the satisfiability of ϕ .

First, we check whether assigning all variables in ϕ to 1 evaluates ϕ to True. If so, the formula is satisfiable, and we have determined its satisfiability. Otherwise, we use Lemma 8 to encode the CNF formula as a neural network f . Next, we compute a subset-minimal δ -sufficient reason by setting $\mathcal{D}$ as the empirical distribution containing only a single data point $\mathbf{0}_n$, following a similar procedure to Lemma 6. Additionally, we set $\delta = 1$ and compute a subset-minimal δ -contrastive reason concerning $\langle f, \mathbf{x} \rangle$.

We will now demonstrate that if any subset-minimal δ contrastive reason obtained for ϕ is satisfiable, then ϕ itself is satisfiable. Conversely, if no subset-minimal δ contrastive reason is obtained, then ϕ is unsatisfiable. The validity of this claim follows a reasoning similar to that provided in Lemma 6. Specifically, the term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$, where the distribution $\mathcal{D}$ considers sampling from a single datapoint, can set the probability to either 0 or 1. Furthermore, since we are searching for a $\delta = 1$ contrastive reason, this is equivalent to asking whether there exists an assignment that changes the classification of f , which corresponds to modifying the assignment of ϕ from False to True. If such an assignment exists, then ϕ is satisfiable. However, if no subset-minimal contrastive reason exists, then no subset of features fixed to zero — when the complementary set is set to ones — evaluates to true. This is equivalent to stating that no assignment evaluates f (and consequently ϕ) to True, implying that ϕ is unsatisfiable.

To extend the proof from neural networks to tree ensembles, we can follow the same procedure outlined in Lemma 6, encoding the CNF formula into an equivalent random forest classifier. Consequently, the proof remains valid for tree ensembles, thereby concluding the proof.

□

K PROOF OF THEOREM 3

Theorem 3. Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $PTIME = NP$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.

Proof. We divide the proof of the theorem into two lemmas, covering both the approximation guarantee and the result on the absence of a bounded approximation.

Lemma 10. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence.*

Proof. We will in fact prove a stronger claim, showing that this holds for *any* model, provided the trivial condition that its inference time is computable in polynomial time, along with one additional mild condition that we will detail later — both of which apply to both our neural network and tree ensemble formalizations.

We begin by noting that, since we are working with empirical distributions, the computation of the global contrastive probability value function:

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

Can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously established in Proposition 2. Since Algorithm 2 performs only a linear number of these polynomial-time queries, its total runtime is therefore polynomial.

Regarding the approximation, the classical work by Wolsey et al. (Wolsey, 1982) established a harmonic-series-based approximation guarantee for monotone non-decreasing submodular functions v with integer values. More generally, their result yields an approximation factor of $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$. We showed in Proposition 2 that the value function v_{con}^g is monotone non-decreasing, and under feature independence, Proposition 4 establishes its submodularity. Combined with the fact that Algorithm 2 runs in polynomial time, this directly yields an approximation guarantee of $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$.

However, we must also show that the expression $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ is both finite and bounded by $\mathcal{O}(\ln(|D|))$, implying that it is effectively constant, due to the assumption of a fixed dataset $|D|$. To ensure this, we begin by confirming that the expression is well-defined and finite. We then proceed to establish the desired bounds. To do so, we introduce a preprocessing step in which we eliminate “redundant” elements—those that could theoretically cause the denominator $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$ to be zero. We begin by formally defining what we mean by redundancy:

Definition 1. *Let $\mathcal{D}$ denote some empirical distribution over a dataset D . Then we say that some feature $i \in [n]$ is redundant with respect to $\mathcal{D}$ if for any pair $\mathbf{x}, \mathbf{z} \in D$ it holds that $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$.*

Here, the notation $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$ indicates that all features in $[n] \setminus \{i\}$ are fixed to their values in $\mathbf{x}$, while feature i is set to its value in $\mathbf{z}$. Notably, this is equivalent to defining:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, the notation $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ indicates that the features in S are fixed to their values in $\mathbf{x}$, while the features in $\bar{S}$ are fixed to their values in $\mathbf{z}$. Thus, in this sense, if we “remove” feature i from the input space and define a new function f' over the reduced space $[n'] := [n] \setminus \{i\}$, then for any input of size n' , the output of f' will exactly match the output of f when applied to the same features $[n] \setminus \{i\}$. This removal can be done in polynomial time for both tree ensembles and neural networks: for neural networks, it involves detaching the corresponding input neuron from the network; for tree ensembles, it involves removing any splits on that feature from all decision trees. Given the empirical nature of the contrastive value function $v_{\text{con}}^g(S)$ — as previously discussed in this proof and in Proposition 2 — we can compute each $v_{\text{con}}^g(\{i\})$ for all $i \in [n]$ by iterating over all pairs $\mathbf{x}, \mathbf{z} \in D$ and checking whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this condition holds, it indicates that feature i is not redundant. Conversely, if the condition fails for all pairs considered during iteration, then i is deemed redundant. Once all redundant features have

been identified with respect to the empirical distribution $\mathcal{D}$, we can remove them and construct an equivalent model f' . As discussed above, this transformation can be performed in polynomial time for both neural networks and decision trees.

To conclude our proof, we observe that $v_{\text{con}}^g(\{i\})$ equals 0 for some empirical distribution $\mathcal{D}$ if and only if feature i is redundant with respect to $\mathcal{D}$. Therefore, once the preprocessing step removes all redundant features from f and we construct the resulting function f' , we ensure that $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$.

We now present the more precise bounds referenced in our proof. Specifically, we demonstrate that the approximation factor is $\mathcal{O}(\ln(|D|))$. Given that the empirical distribution $\mathcal{D}$ is fixed, this yields a constant approximation. While tighter bounds may be achievable, our goal here is solely to establish that the bound is *constant* — a property that will later sharply contrast with the local explainability setting, where no bounded approximation exists.

More specifically, we know that the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is, by the definition of empirical distributions, at least $\frac{1}{|D|}$. Therefore, the definition of $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ (i.e., the value function v_{con}^g) is at least $\frac{1}{|D|^2}$. In particular, this also implies that:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|D|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right) \leq \ln(|D|^2) = \mathcal{O}(\ln(|D|)) \quad (71)$$

Which concludes our proof.

□

Lemma 11. *Unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a cardinally minimal local δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$ where f is either a neural network or a tree ensemble, even when $|D| = 1$.*

Proof. Assume, for contradiction, that there exists a bounded approximation algorithm for computing a cardinally minimal local δ -contrastive reason for some $\langle f, \mathbf{x} \rangle$, where f is a neural network or a tree ensemble—even when $|D| = 1$. However, Lemma 9 establishes that even with a single baseline $\mathbf{z} = 0_n$ (i.e., when the entire dataset is just one instance), deciding whether a contrastive reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Therefore, if a polynomial-time algorithm could yield a bounded approximation for a cardinally minimal contrastive reason, it would contradict this result, as such an approximation would implicitly decide the existence of a contrastive explanation.

□

L PROOF OF THEOREM 4

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-kf} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We divide the proof into two lemmas: one establishing the approximation guarantee for the global case, and the other demonstrating the intractability of the local case.

Lemma 12. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-kf} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence.*

Proof. The proof will follow a similar approach to that of Lemma 10, where we showed that for both neural networks and tree ensembles, Algorithm 2 achieves an approximation factor of $\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right)$. After applying the preprocessing step, this ratio is guaranteed to be finite and bounded by $\mathcal{O}(\ln(|D|))$.

Similar to Lemma 10, we will again prove a stronger claim — namely, that the result holds for *any* model, assuming the trivial condition that its inference time is polynomially computable, and the additional condition concerning the removal of redundant features, as described in Lemma 10. As discussed there, both conditions are satisfied by our neural network and tree ensemble formalizations.

Here as well, since we are working with empirical distributions, the computation of the global sufficient probability value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as previously shown in both Lemma 10 and Proposition 2. Since Algorithm 2 makes only a linear number of such polynomial-time queries, its overall runtime is polynomial.

Unlike Lemma 10, where the approximation relied on the result by Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the setting here requires a different condition due to the supermodular nature of the function. Specifically, Shi et al. (Shi et al., 2021) provided an approximation guarantee of $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$.

We established in Proposition 2 that the value function v_{suff}^g is monotone non-decreasing, and under the assumption of feature independence, Proposition 3 further shows that it is supermodular. Given that Algorithm 2 runs in polynomial time, this leads directly to an approximation guarantee of $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$, where $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)}$.

To show that this expression is both bounded and constant, we follow the same preprocessing step as in Lemma 10, where we remove all redundant features from f — a process that, as previously explained, can be carried out in polynomial time for both neural networks and tree ensembles. As before, a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. This preprocessing yields a new function f' that behaves identically to f over the remaining features and ensures that, for every i , both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

In order for us to show that this expression is both bounded and constant, similarly to Lemma 10 we will perform the exact same preprocessing phase (which, as we explained there, can be performed in polynomial time both for neural networks as well as tree ensembles) as before where we remove all redundant features from f . We note that here too, it holds that a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. Hence, this preprocessing phase will give us a new function f' for which it both holds that it is equivalent to the behaviour of f for all remaining features and also satisfies that for any i it holds that both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

Now, following the same reasoning as in Lemma 10, where the probability term is lower bounded by $\frac{1}{|D|}$ and its expected value by $\frac{1}{|D|^2}$, we obtain that $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|D|^2}$ and $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|D|^2}$. This implies, as in Lemma 10, that the first term is lower bounded by $\mathcal{O}(\ln(|D|))$, and for the second term we derive the following:

Since v_{suff}^g is supermodular (Proposition 3), we have that $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$. It follows that $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$. This implies: $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |D|^2$, and therefore, $1 - |D|^2 \leq k^f \leq 0$. This yields $\frac{1}{1-k^f} \leq |D|^2$, demonstrating that the overall approximation bound based on the one established by Shi et al. is constant. While this bound may be significantly smaller in practice, our goal here is simply to show that it remains constant — unlike in the local setting, where no bounded approximation is achievable.

□

Lemma 13. *Unless $PTIME=NP$, there is no bounded approximation for computing a local cardinality minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The proof follows a similar approach to Lemma 11. Suppose, for contradiction, that there exists a polynomial-time algorithm that provides a bounded approximation to a cardinally minimal local δ -sufficient reason for some $\langle f, \mathbf{x} \rangle$, where f is either a neural network or a tree ensemble—even when $|\mathbf{D}| = 1$. Yet, Lemma 6 shows that even in the extreme case where the dataset consists of a single baseline $\mathbf{z} = \mathbf{0}_n$, deciding whether a sufficient reason exists is NP-hard unless PTIME = NP. Hence, the existence of such an approximation algorithm would contradict this hardness result, as it would entail the ability to decide the existence of a sufficient reason.

□

M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

Proposition 1. Assume a neural network $f : \mathbb{F} \rightarrow [c]$ with ReLU activations, some dataset $\mathbf{D}$, such that $\mathcal{D}$ is the empirical distribution defined over $\mathbf{D}$, and some threshold $\delta \in \mathbb{Q}$. Then unless $NP \subseteq DTIME(n^{\mathcal{O}(\log(\log(n)))})$, there does not exist a polynomial-time $\mathcal{O}(\ln(|\mathbf{D}|) - \epsilon)$ approximation algorithm for computing a cardinally minimal global contrastive reason for f concerning $\mathcal{D}$, for any fixed $\epsilon > 0$.

Proof. The proof proceeds via a polynomial-time approximation-preserving reduction from a specific variant of Set Cover that is, similarly to regular Set Cover, provably hard to approximate within the factor given in the proposition (Vazirani, 2001).

Partial Set Cover:

Input: A universe $U := \{u_1, \dots, u_n\}$, a collection of sets $S := \{S_1, \dots, S_m\}$ such that for all j $S_j \subseteq U$, and a coverage threshold $\delta \in \{0, 1, \dots, n\}$.

Output: A minimum-size subset $C \subseteq S$ such that $|\cup_{S_j \in C} S_j| \geq \delta$.

We reduce this problem to that of computing a *global*, cardinally-minimal contrastive reason S for a neural network f with respect to a distribution $\mathcal{D}$ defined over a dataset $\mathbf{D}$, and some threshold δ' . We begin by recalling the definition of a global contrastive reason and then extend it to the setting where the explanation is evaluated over an empirical dataset $\mathbf{D}$. The definition is as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More specifically, when we assume an empirical distribution over $\mathbf{D}$, this implies that:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) \end{aligned} \quad (74)$$

where we denote $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$. We now construct a ReLU neural network f . Before doing so, we specify the functional form that we intend the constructed neural network f to implement. For each set S_j , we introduce two vectors $\mathbf{x}^j$ and $\mathbf{y}^j$, both belonging to $\mathbb{Z}^n$ and drawn from the dataset $\mathbf{D}$, such that they agree on all coordinates outside S_j ; that is, they share identical feature values on the complement $\bar{S}_j$. Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and we assign strictly different values to the features in S_j by selecting them from two distinct vectors $\mathbf{z}$ and $\mathbf{z}'$ such that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = \mathbf{y}^j \quad (76)$$

We aim to construct f such that it satisfies the following:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{S_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{S_j}^j; \mathbf{z}_{S_j}^{\prime j}) = f(\mathbf{y}_{S_j}^j; \mathbf{z}_{S_j}^{\prime j}) = f(\mathbf{y}^j) \quad (77)$$

Let us now assume that both $\mathbf{x}^j$ and $\mathbf{y}^j$ occur *uniquely* in the dataset $\mathbf{D}$ across all $[n]$ coordinates — that is, no other vector in $\mathbf{D}$ matches either of them on any coordinate. Under this assumption, we obtain the following relation:

$$\mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{S_j} = \mathbf{x}_{S_j})] = \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{S_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{|\mathbf{D}_{S_j}(\mathbf{z}, \mathbf{x})|} \right) = \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \quad (78)$$

Here, the factor of 2 appearing in both the numerator and denominator arises from the fact that only two vector pairs — $(\mathbf{x}^j, \mathbf{z}^j)$ and $(\mathbf{z}^j, \mathbf{x}^j)$ — contribute to the respective sums.

We repeat this construction for each S_i in the set-cover instance, each time choosing two *distinct* vectors $\mathbf{x}^i$ and $\mathbf{y}^i$ that differ on *every* coordinate in $[n]$. In total, this yields $2m$ vectors that form the dataset $\mathbf{D}$ from which we construct f .

We then construct a neural network f that simulates this function. This is achieved using a known result on *neural network memorization* (Vardi et al., 2022), which guarantees that a ReLU network capable of representing any finite set of input–output pairs can be built in polynomial time. Using this result, we construct f accordingly, and set $\delta' := \frac{\delta}{|\mathbf{D}|}$.

We now argue that a subset S_j is a global contrastive reason for f under tolerance δ' if and only if C is a partial set cover for U under tolerance δ . By Equation 78, each constructed pair of vectors $(\mathbf{x}^j, \mathbf{y}^j)$ contributes exactly $\frac{1}{|\mathbf{D}|}$ to the value term. Moreover, because each S_j uses its own distinct pair of vectors in $\mathbb{Z}^n$, there is no overlap between these pairs. Consequently, the coverage achieved by any set cover C corresponds exactly to the accumulated contribution of the selected subsets in the input domain of f . This completes the reduction.

□

N TRACTABILITY RESULTS FOR ORTHOGONAL DNFs

Lemma 14. *If f is an orthogonal DNFs and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time.*

Proof. We begin by defining orthogonal DNFs (Crama & Hammer, 2011) before presenting the full proof.

Definition 2 (Orthogonal DNFs). *A Boolean function φ is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms $T_1 \vee T_2 \vee \dots \vee T_m$, such that for every pair of distinct terms T_i and T_j ($i \neq j$):*

$$T_i \wedge T_j \models \perp$$

This means that no single variable assignment can satisfy more than one term simultaneously.

The proof mirrors the argument used in Theorem 1, relying on the fact that v_{suff}^g can be computed in polynomial time for any set S .

For a given set S , let $\mathcal{C}^S T_i$ be the constraints that the term T_i imposes on the variables in S . We enumerate all pairs of terms (T_i, T_j) . Any pair whose constraints on S are *inconsistent* (i.e., $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$) is ignored, since it contributes zero to $v^g \text{suff}$.

2214 For each *consistent* pair of terms (T_i, T_j) , we evaluate $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$, where $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$.
 2215 The contribution of this pair to v_{suff}^g is then
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$$\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

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 2219 Under feature independence, these probabilities factor as $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$. Since there are at
 2220 most m^2 such term pairs, the value of v_{suff}^g can be computed in polynomial time under an independent
 2221 distribution $\mathcal{D}$. $\square$
 2222

2223 O DISCLOSURE: USAGE OF LLMs

2224
 2225 An LLM was used exclusively as a writing assistant to refine grammar and typos and improve clarity.
 2226 It did not contribute to the generation of research ideas, study design, data analysis, or interpretation
 2227 of results, all of which were carried out solely by the authors.
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